

COURSE TITLE : BASIC MEDICAL SCIENCE
COURSE CODE : 3241
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Introduction to Human Anatomy and Bioelectricity	15
2	Cardiovascular and Nervous Systems	15
3	Respiration, Haemodialysis, Vision and Audition	15
4	Vision, Audition and Human Digestive System	15
	TOTAL	60

Course General Outcome:

MODULE	G.O	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE
1	1	To understand human anatomy.
	2	To comprehend the generation of bio-potentials in cells.
	3	To understand the anatomy and physiology of cardiovascular system
2	4	To comprehend the electrical activity of heart
	5	To understand the anatomy and physiology of nervous system and discuss on temperature regulation
3	6	To understand the respiratory system
	7	To understand the excretory system
4	8	To know the process of vision and audition
	9	To understand digestive system

On completion of the study the student will be able:

MODULE I INTRODUCTION TO HUMAN ANATOMY AND BIOELECTRICITY

1.1.0 To understand human anatomy.

- 1.1.1 To define anatomy, physiology, biochemistry, biophysics.
- 1.1.2 To describe the structure of human cell.
- 1.1.3 To classify elementary tissues of the body, types and their functions.
- 1.1.4 To explain different biological sensors and their mechanism of action
- 1.1.5 To list various types of receptors in human body
- 1.1.6 Explain sensors for smell, sound, vision, osmolality & taste

1.2.0 To comprehend the generation of bio-potentials in cells.

- 1.2.1 To explain bio-electric potentials.
- 1.2.2 To describe the transmission of impulse through nerve fibers.
- 1.2.3 To differentiate between monophasic and biphasic recording.
- 1.2.4 To identify different types of bio-potential electrodes for recording ECG
- 1.2.5 To identify different types of EEG electrodes.
- 1.2.6 To identify different types of EMG electrodes

MODULE II CARDIOVASCULAR AND NERVOUS SYSTEM

2.1.0 To understand the anatomy and physiology of cardiovascular system

- 2.1.1 To describe the cardiovascular system
- 2.1.2 To describe the structure of heart and the circulation of blood.
- 2.1.3 To define cardiac cycle, blood pressure, cardiac output.
- 2.1.4 To explain Electromyography.

2.2.0 To comprehend the electrical activity of heart.

- 2.2.1 To explain the electrical activity of the heart.
- 2.2.2 To explain the ECG waveform and relate it with the mechanical events of the heart.
- 2.2.3 To classify different types of arrhythmias and cardiac myopathies.

2.3.0 To understand the anatomy and physiology of nervous system and temperature regulation.

- 2.3.1 To describe the nervous system.
- 2.3.2 To describe the general organization of brain.
- 2.3.3 To explain the electrical activity of brain.
- 2.3.4 To describe the temperature regulation of human body.
- 2.3.5 To list heat and cold applications in medicine.

MODULE III RESPIRATORY AND EXCRETORY SYSTEM.

3.1.0 To understand the respiratory system and excretory system

- 3.1.1 To describe the anatomy of respiratory system.
- 3.1.2 To explain the physiology of respiration
- 3.1.3 To explain mechanism of ventilation.
- 3.1.4 To define different lung volumes and capacities.

3.2.0 To understand the excretory system

- 3.2.1 To explain excretory system.
- 3.2.2 To describe the structure and function of nephron.
- 3.2.3 To describe haemodialysis.

MODULE IV VISION, AUDITION AND HUMAN DIGESTIVE SYSTEM

4.1.0 To know the process of vision and audition.

- 4.1.1 To explain the structure of eye.
- 4.1.2 To describe the visual process.
- 4.1.3 To explain the structure of ear.
- 4.1.4 To describe process of hearing
- 4.1.5 To describe equilibrium balance.

4.2.0 To understand the digestive system.

- 4.2.1 To describe the anatomy of human digestive system.
- 4.2.2 To illustrate the process of digestion.

CONTENTS

MODULE I INTRODUCTION TO HUMAN ANATOMY

Definition of anatomy, physiology, biochemistry, biophysics – Structure of human cell - Elementary tissues of the body- types and their functions - Biological sensors - mechanism of action - receptors in human body - sensors for smell, sound, vision, osmolality & taste.

Generation of bio-potentials in cells

Origin of bio-electric potentials - Transmission of impulse through nerve fibers - Monophasic and biphasic recording - Types of ECG, EEG and EMG electrodes

MODULE II CARDIOVASCULAR SYSTEM

Structure of heart - circulation of blood – Definitions of cardiac cycle, blood pressure, cardiac output – Electromyography - Electrical activity of the heart – Morphology of ECG waveform – Relation of mechanical events of the heart with electrical activity – Arrhythmias - cardiac myopathies.

Nervous system and temperature regulation

Structure of nervous system - Organization of brain- Electrical activity of brain - Temperature regulation of human body - Applications of heat and cold in medicine

MODULE III RESPIRATION SYSTEM

Anatomy of respiratory system - Physiology of respiration - Mechanism of ventilation – Definition of lung volumes and capacities

Excretory system

Anatomy of Excretory system - structure and function of nephron – haemodialysis

MODULE IV VISION AND AUDITION

Structure of eye - visual process - structure of ear - Process of hearing - Equilibrium and balance
Human Digestive System

Anatomy of human digestive system - Process of digestion - metabolism of carbohydrates, fats and proteins.

TEXT BOOKS

1. Anatomy and Physiology for Nurses - Evelyn C Pearce-Jaypee Brothers Medical Publishers (P) Ltd
2. Introduction to Biomedical Instrumentation - Leslie Cromwell-Prentice Hall-2011

REFERENCES

1. A textbook of Medical Physiology - Artur C Guyton & Hall-Prism Books (P) Ltd & W B Saunders Company-2006
2. Hand book of Biomedical Instrumentation - R S Khandpur-Tata McGraw Hill
3. Applied Physiology - Samson Wright, Cyril A Keele, Oxford University Press 1990